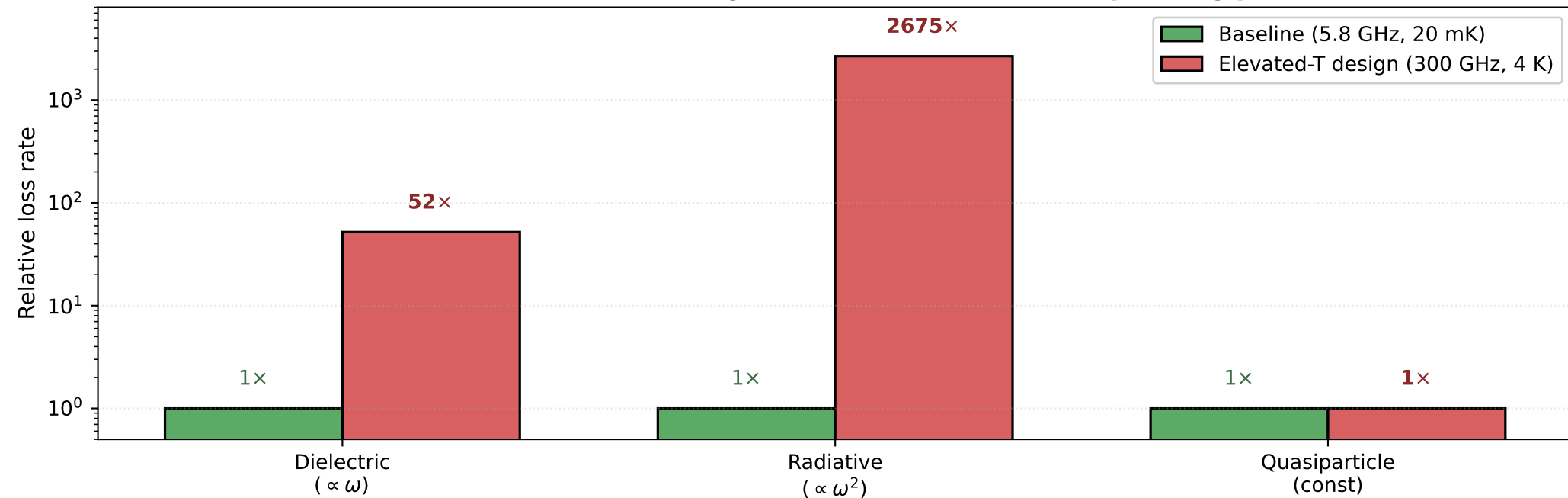
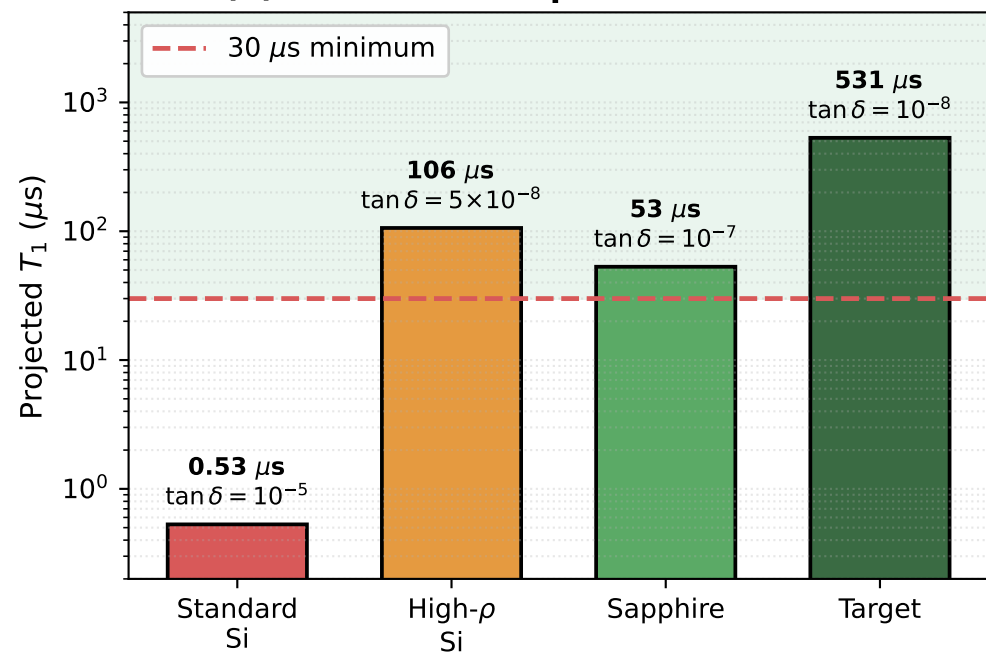


(a) Loss-mechanism scaling: baseline vs. elevated-T operating point



(b) Substrate comparison at 300 GHz



(c) Improvement pathways at fixed 4 K

